

HAMR Temperature and Field Map (THMap) Measurements with Minor Loops

Pierre-Olivier Jubert¹, Toan Tang¹, and Richard Brockie¹

¹ WD, San Jose, CA 95119, USA, Pierre-Olivier.Jubert@wdc.com

Details of the heat-assisted magnetic recording (HAMR) media switching processes and distributions are explored using an extension of the temperature and field map (THMap) measurements. Using DC-noise measurements in conventional recording systems, the new method extracts the probability of magnetic clusters switching as a function of applied temperature, applied field, and varying initial magnetization state, thus investigating temperature and field remanent minor loops. An analytical noise model is presented that reproduces the experimental steps and fits the experimental data very well. The measurements allow a refinement of the understanding of HAMR media switching mechanisms at the temperatures and time scales relevant to HAMR recording.

Index Terms— Heat Assisted Magnetic Recording (HAMR), magnetic media, magnetic recording, measurement

I. INTRODUCTION

HEAT-ASSISTED magnetic recording (HAMR) technology has now been introduced in hard-disk drives [1],[2]. HAMR media consist of high-anisotropy FePt grains that are written by the writer field while the grains cool just below their Curie temperature, T_c . The localized heating is provided by the HAMR head near-field transducer excited by an embedded laser diode. High areal density results from high temperature gradients and narrow media switching distributions.

Understanding the FePt grains' switching process and distributions at recording time scales is important for the development of HAMR media. To characterize the media properties, a method based on reverse-DC noise measurements, named THMap, has been proposed [3],[4].

In this work, an extension of the measurements with temperature sweeps that start from partially erased states is presented. These are equivalent to minor remanent loops. The THMap minor loop method is described first, and typical measurement results are presented. An analytical noise model of THMap minor loop is derived and compared to the experimental data.

II. THMAP MINOR LOOP MEASUREMENTS

The proposed measurement method is based on DC noise measurements in HAMR recording systems. First, the medium is initialized in a DC saturated state by band writing at a large writer current, I_{w1} , and a laser power large enough for the medium temperature, T_{peak1} , to be well above T_c . Band writing ensures that all the grains in the probed area experience the same temperature and the same field. Second, the area is overwritten over the same band at a peak temperature T_{w2} , with a field of amplitude and direction defined by the writer current I_{w2} , which can be positive or negative. Under these conditions, a proportion p_2 of the magnetic clusters may change their orientation compared to the initially saturated state. For I_{w2} of opposite polarity to I_{w1} , measuring the readback noise at varying T_{peak2} and I_{w2} give the conventional THMap, from which laser power can be converted into T_{peak} values and which provides switching temperature distributions at the different applied fields [3],[4]. In this work, after step 2, a third band-write step

is performed at a peak temperature $T_{peak,3}$ and an applied field corresponding to the writer current I_{w3} . The readback noise is then captured in the center of the written band as a function of $(T_{peak2}, I_{w2}, T_{peak3}, I_{w3})$.

Fig. 1 shows the evolution of the readback noise power with increasing T_{peak3} for selected I_{w3} of positive and negative polarities, after the medium was prepared in different magnetization states at (T_{peak2}, I_{w2}) . This allows exploration of the evolution of the probability of switching with increasing temperature and field starting from different magnetization states, which is equivalent to remanent minor loops.

For example, Fig. 1a shows the evolution of the noise power,

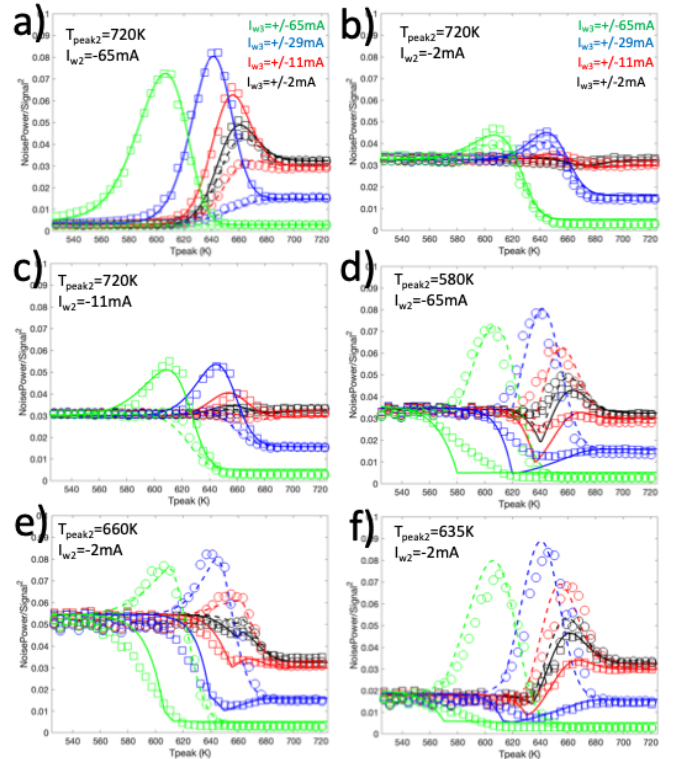


Fig. 1. THMap minor loops for selected initial magnetization states: experimental data (symbols) and analytical model results (lines). Colors correspond to different applied fields. Squares (circles) correspond to fields favoring switching in the direction opposite (same) to the initial DC state.

which relates to the proportion of up/down magnetic clusters, as a function of increasing temperature, for positive and negative fields, starting from a negatively DC-saturated state. The initial states for Fig. 1b and 1c are thermally erased states set at $T_{peak2} \gg T_c$ and $I_{w2} = -2\text{mA}$ and -11mA , respectively. For Fig. 1d, $T_{peak2} = 580\text{K}$, which is lower than the mean writing temperature $\langle T_w \rangle$ at $I_{w2} = -I_{w1}$, such that only magnetic clusters with a low switching temperature were switched before applying T_{peak3} and I_{w3} . In Fig. 1e and 1f, the initial magnetization state is partially switched at $T_{peak2} \sim T_c = 660\text{K}$ and $T_{peak2} = 635\text{K}$, respectively, with $I_{w2} = -2\text{mA}$.

The THMap minor loops curves provide additional characterization of the switching behavior of the HAMR medium as a function of the temperature and field history, which can be captured by replicating the experimental data.

III. NOISE ANALYTICAL MODEL FOR THMAP MINOR LOOPS

Analytical expressions for the DC signal variance of conventional THMap measurements have been reported previously [3],[4]. These expressions are extended here to reproduce the probability of magnetic clusters switching as a function of the three-steps temperature and applied field history.

The DC noise power after the three writes is given by:

$$NP(T_{peak3}, I_{w3}, T_{peak2}, I_{w2}) = \left(\frac{S_{sat}^2}{R_{sat}^2} \right) \cdot \frac{1}{N_{eff}} R^2 \left(\frac{1}{R} - (2p_3 - 1)^2 \right) \quad (1)$$

S_{sat} is the DC signal amplitude at high field and high peak temperature. N_{eff} is the effective number of independent magnetic clusters under the reader. R_{sat} is the proportion of out-of-plane magnetic clusters at saturation.

The magnetic clusters' temperature switching distribution is assumed to be skew normal with $T_{loc}(I_w)$ the location of the distribution, $\sigma_{Tw}(I_w)$ the scale of the distribution, and $\gamma(I_w)$ the shape parameter. At each writer current, the cumulative distribution function of the temperature switching distribution is given by:

$$CDF_p(T_{peak}, I_w) = \left(0.5 + 0.5 \cdot \text{erf} \left(\frac{T_{peak} - T_{loc}(I_w)}{\sqrt{2} \cdot \sigma_{Tw}(I_w)} \right) \right) - 2 \cdot T \left(\frac{T_{peak} - T_{loc}(I_w)}{\sqrt{2} \cdot \sigma_{Tw}(I_w)}, \gamma_{Tw}(I_w) \right) \quad (2)$$

where $T(h, \gamma)$ is Owen's T function.

The model assumes that an out-of-plane magnetic cluster that experiences a temperature T_{peak} larger than its switching temperature T_w aligns opposite to the initial magnetization direction with a probability α . α varies from 0.5 to 1 with I_w changing from 0 to $-I_{w1}$ and symmetrically from 0.5 to 0 with I_w changing from 0 to $+I_{w1}$. Considering the switching probabilities at (T_{peak2}, I_{w2}) and then at (T_{peak3}, I_{w3}) , leads to:

$$p_3(T_{peak3}, I_{w3}, T_{peak2}, I_{w2}) = \alpha(I_{w3}) \cdot CDF_p(T_{peak3}, I_{w3}) + \alpha(I_{w2}) \cdot \max \left(0, CDF_p(T_{peak2}, I_{w2}) - CDF_p(T_{peak3}, I_{w3}) \right) \quad (3)$$

As observed in Fig. 1, the noise power as a function of temperature does not peak at the same level for the different writer currents and initial magnetization states. This would be the expectation for fixed N_{eff} and R values: the noise power peak when T_{peak3} is such that $p_3 = 0.5$ is $NP = S_{sat}^2 \cdot R / N_{eff} \cdot R_{sat}^2$. To match

the experimental data, the cluster size via N_{eff} and/or the proportion of out-of-plane grains R need to be adjusted as a function of temperature and field. The change of R with temperature and field may arise from the creation of multi-domain states at high temperatures, low fields [5]-[8]. In the present analytical model, we enable R to change with temperature and field as follows:

$$R_3(T_{peak3}, I_{w3}, T_{peak2}, I_{w2}) = R_{sat} \left[1 - \beta(I_{w3}) \cdot CDF_p(T_{peak3} - \Delta T_R(I_{w3})) - \beta(I_{w2}) \cdot \max \left(0, CDF_p(T_{peak2} - \Delta T_R(I_{w3})) - CDF_p(T_{peak3}) \right) \right] \quad (4)$$

Fitting the experimental data from Fig. 1a and 1b provides the probability of aligning to the field $\alpha(I_w)$, the magnetic cluster switching distribution parameters $T_{loc}(I_w)$, $\sigma_{Tw}(I_w)$, $\gamma_{Tw}(I_w)$, and the probability of decreasing the proportion of out-of-plane clusters $\beta(I_w)$ and $\Delta T_R(I_w)$. β decreases from 0.5 at $I_w = 0$ to 0 at $I_w > 25\text{mA}$. Using these parameters and equations (1) to (4) allows replicating all the minor loop THMap curves of Fig. 1 very well.

IV. CONCLUSIONS

THMap minor loops are presented, characterizing the evolution of the medium DC noise with varying applied temperatures and fields, after preparing the medium in different magnetization states. Details on the switching mechanisms and distributions of the HAMR medium are revealed by fitting the experimental data. An analytical noise model is proposed that shows very good agreement with the experimental data. It requires considering that, at high temperatures and low fields, the proportion of out-of-plane magnetic clusters varies. Reproducing the experimental THMap minor loops with micromagnetic simulations would be of interest to validate the analytical model assumptions and confirm the intrinsic properties and switching dynamics of the FePt grains at high temperatures.

REFERENCES

- [1] M. H. Kryder, E. C. Gage, T. W. McDaniel, W. A. Challener, R. E. Rottmayer, G. Ju, Y.-T. Hsia, M. F. Erden, "Heat assisted magnetic recording," *Proc. IEEE*, vol. 96, no. 11, pp. 1810-1835, Nov. 2008.
- [2] S. Granz, C. Rea, G. Ju, P. Czoschke and S. Hernandez, "Heat assisted magnetic recording areal density dependence on writer current for conventional and shingled magnetic recording", *Proc. IEEE 34th Magn. Recording Conf. (TMRC)*, pp. 1-2, Jul. 2023.
- [3] P.-O. Jubert et al., "Spin-stand measurements to extract the switching distributions of heat-assisted magnetic-recording media", *IEEE Trans. Magn.*, vol. 59, no. 3, Mar. 2023.
- [4] P.-O. Jubert, "Analytical DC Noise Models for Granular Perpendicular Media," in *IEEE Transactions on Magnetics*, vol. 61, no. 11, pp. 1-13, Nov. 2025, Art no. 3000613
- [5] J.-G. Zhu and Y. Yan, "Incoherent Magnetic Switching of L1₀ FePt Grains," in *IEEE Transactions on Magnetics*, vol. 57, no. 3, pp. 1-9, March 2021
- [6] J.-G. Zhu, P.-O. Jubert, A. Moser, R. Brockie and R. Shi, "The Enabling Mechanism for High Track Density in HAMR," in *IEEE Transactions on Magnetics*, vol. 61, no. 4, pp. 1-9, April 2025
- [7] L. Xu, V. Le, T. Olson and K. Tanahashi, "The Computational Analysis of HAMR Media Noise with a Multilayer Micromagnetic Media Model," in *IEEE Transactions on Magnetics*, vol. 62, no. 3, pp. 1-6, March 2026
- [8] J.-G. Zhu, "Single-Pass Three-Level Recording with Granular FePt-L1₀ Medium," in *IEEE Transactions on Magnetics*, vol. 62, no. 3, pp. 1-8, March 2026